

The Boundary Card

ChatGPT edition / Print and post

Post beside the bench; read before every investigation

LEARNER DOMAIN

Only isolated, current-limited battery circuits at **12 V DC or less**, using the parts and limits printed in the investigation. Disconnect before rewiring. This conservative course rule controls exposure; it does not mean everything below 12 V is harmless.

Green domain — learner builds, adult checks

- AA-cell holders or the specifically listed small battery source
- intact insulated clip leads, listed resistors, LEDs, small coils
- hand-moved magnets, compass, passive sensors
- resistance/continuity with every source removed
- voltage/current measurement only exactly as the investigation specifies

Black-line domain — stop; outside this course

Household power	No outlet experiments, socket probes, mains wiring, power strips, lamps, appliances, extension-cord repair, or opened power supplies.
High voltage	No Tesla coils, ignition coils, microwave parts, flyback transformers, stun devices, spark gaps, CRTs, neon transformers, or arcs.
Stored/fault energy	No large capacitor banks, vehicle/lead-acid batteries, lithium packs without an approved protected holder, or improvised high-current supplies. Low voltage can still start a fire or cause burns.
Unknown apparatus	No unlabelled adapter, salvaged supply, damaged cell, frayed lead, swollen battery, or mystery component.

Adult ownership is not qualification

The facilitator owns the kit, checks the planned circuit, and makes every stop decision. That role does not qualify anyone to work on exposed mains or high voltage. OSHA confines exposed energized-parts work to qualified persons and normally requires exposed parts to be deenergized and verified with test equipment. This course contains no procedure for that work.

Stop signals

Disconnect without touching a hot or damaged part. Tell the adult.

- warmth, odor, smoke, sparking
- swelling, leak, discoloration
- damaged insulation or loose metal
- meter over-range or unstable display
- circuit differs from the diagram
- someone proposes “just try it”

FIVE CHECKS BEFORE CONNECT

☐ Listed battery source only ☐ No direct source short ☐ Meter mode, range and jacks correct ☐ Adult checked diagram against build ☐ Everyone knows the disconnect point

ELECTRICAL EMERGENCY

Do not touch a person who may still be energized. Disconnect power without entering the current path and call emergency services. Do not improvise rescue or rely on a switch/indicator as proof that hazardous equipment is deenergized.

Evidence: OSHA S01–S03 and S02; CPSC S08–S09. See the provider-neutral electricity source register.

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